TUSHAR SINGH

AI Engineer | Full-Stack Developer

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OBJECTIVE

Builds AI-powered products and automation systems that take real work off people's plates — from multilingual voice assistants and computer-vision biometrics to full-stack production web apps. Has shipped 15+ automation workflows and 8+ production web applications for clients, consistently cutting manual effort by up to 60% while keeping systems reliable and easy to maintain. Comfortable owning a project end-to-end: talking to the client, designing the system, writing the code, and deploying it. Immediate joiner, open to relocation.

EXPERIENCE

Freelance AI Developer — Remote *Jan 2025 – May 2026*

- Built 15+ AI automation workflows using Python, n8n, and REST APIs, cutting client manual effort by 60% across 10+ engagements.
- Delivered 8+ full-stack production web apps (Next.js, React, Django, Supabase, MySQL), maintaining 99% uptime for client systems.
- Integrated 10+ APIs (OpenAI, Gemini, Groq, WhatsApp Business, Stripe, Razorpay) to automate customer interactions for 500+ monthly users.

Data Analyst Intern — AB Infotech Solutions, Pune *Jun 2025 – Sep 2025*

- Analyzed 50,000+ business records using Python and SQL, cleaning and exploring the data to improve downstream reporting reliability.
- Built 5+ Power BI dashboards, surfacing 12+ trends that informed 3 executive decisions.
- Developed an interactive Streamlit application combining data analysis and dashboard visuals, so the team could explore business data without writing code.
- Automated reporting workflows, cutting manual effort by 40%.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript (ES6+), TypeScript, Java
- **AI & Generative AI:** LLMs, AI Agents, Prompt Engineering, RAG, MCP Concepts, OpenAI, Gemini, Claude, Groq, OpenRouter, LangChain, LangGraph, CrewAI, ChromaDB
- **Backend & APIs:** FastAPI, Django, Django REST Framework, Node.js, Next.js (App Router), REST APIs, Webhooks, WebSockets, Auth, ORM
- **Frontend:** React.js, Next.js, TypeScript, Tailwind CSS, Bootstrap, Responsive UI
- **Databases:** PostgreSQL, Supabase, MySQL, SQLite, MongoDB
- **DevOps & Tools:** Git, GitHub, Docker, n8n, Make.com, Cloudflare R2, Upstash Redis, Postman, Figma
- **Cloud & Payments:** Google Cloud Console (OAuth 2.0 authentication & authorization, client ID/secret, access/refresh tokens), AWS (Basic), Render, Stripe, Razorpay

KEY PROJECTS

TASQ-ONE — AI-Powered Multi-Tenant Work OS *Live Demo*

Tech: *Next.js 14, TypeScript, Supabase, Groq LLM, Tailwind CSS, Upstash Redis, Cloudflare R2, Slack Webhooks, Google OAuth 2.0, Docker*

- Built an enterprise-grade multi-tenant task/sprint management platform with AI-powered deliverable decomposition (Groq Llama 3.3 70B, sub-second inference), 3-tier RBAC, PostgreSQL Row-Level Security, dependency DAG blocking, and automated Slack alerts. Set up Google Cloud Console OAuth 2.0 (client ID/secret, redirect URIs, token handling) for secure sign-in and authorization. Deployed via Docker on Render; 30/30 tests passing.

Live: tasq-one.onrender.com

Saathi Bot — Multi-Tenant RAG Chatbot SaaS *Live Demo*

Tech: *Next.js 16, React 19, TypeScript, Supabase (PostgreSQL, pgvector, pgmq), Groq Llama 3.3 70B, Gemini 2.5, Google OAuth 2.0, Tailwind CSS, Vercel*

- Built a B2B SaaS that ingests websites, sitemaps, PDFs and DOCX into a pgvector knowledge base and deploys a grounded, Hinglish-capable chatbot via a single **<script>** tag (~24 KB Shadow DOM widget). Added a similarity-based confidence gate that skips LLM calls when no match exists (cuts token spend, prevents hallucinated answers), in-chat lead booking with Google Sheets sync via Google Cloud Console OAuth 2.0 (client ID/secret, access/refresh tokens, scoped authorization), a public REST API with SSE streaming, and Row-Level-Security tenant isolation verified by automated tests (0 cross-tenant leaks).

Live: saathi-bot.vercel.app

TakeOpinion — Medical Tourism Platform *Live Demo*

Tech: *Django 5.2, Django REST Framework, PostgreSQL, Bootstrap 5, Celery, Redis, Stripe, Razorpay*

- Built a full-stack medical tourism platform connecting patients with doctors, hospitals, and treatments — covering treatment catalog, doctor/hospital directories, appointment booking with Google Meet integration, hotel accommodation booking, and secure payment processing via Stripe and Razorpay. Implemented blog CMS, review system, and REST search APIs; deployed on Render with Gunicorn and Whitenoise.

Live: takeopinionclientprojectss.onrender.com

ShieldCaptcha — Zero-Cookie, Self-Hostable Bot Defense Platform *Live Demo*

Tech: *Next.js 16, React 19, TypeScript, Node.js, FastAPI, Docker, Nginx*

- Built an open-source, privacy-preserving CAPTCHA alternative with no tracking cookies: client-side SHA-256 proof-of-work in a Web Worker, pointer-kinematics analysis (curvature, jitter, acceleration, micro-tremor) to flag headless automation, and single-use HMAC-SHA256 tokens with 300s expiry for replay protection. Shipped dual verification engines (zero-dependency Node.js and FastAPI), an <18 KB web component, a React wrapper, and Docker/Nginx deployment.

Live: shieldcaptcha.vercel.app

Riya AI — Multilingual Voice Assistant & Autonomous Desktop Agent

Tech: *Python, Groq Llama 3.3 70B, Vosk, PyQt5, OpenCV, dlib, SQLAlchemy, gTTS*

- Built an enterprise-grade multilingual (Hindi/Hinglish/English) voice assistant with offline wake-word detection, face-biometric authentication, and direct OS/hardware automation (volume, brightness, app launch, real-time voice dictation). Powered by sub-second Groq Llama 3.3 70B intent extraction with structured JSON entity parsing, voice-driven email dispatch via SMTP, and a PyQt5 holographic HUD interface.

Repo: github.com/Tusharsinghoffical/Riya-Assistant

FaceVision AI — Deep Biometric Demographics & Recognition Platform

Tech: *Python, DeepFace, OpenCV, TensorFlow, Scikit-Learn, Streamlit, Plotly*

- Built a computer-vision biometrics platform delivering age/gender/emotion/ethnicity estimation, 68-point 3D facial mesh with Euler pose (pitch/yaw/roll), 1:1 face verification and 1:N gallery recognition against a 128-d embedding vector database, plus PCA/t-SNE/DBSCAN latent-space exploration for group and batch analysis.

Repo: github.com/Tusharsinghoffical/Face-Recognition-Analysis

EDUCATION

Master of Computer Applications (MCA) — MIT ADT University, Pune *Oct 2024 – Apr 2026 | 79%*

Bachelor of Computer Applications (BCA) — Institute of Technology & Science, New Delhi *Oct 2021 – Mar 2024 | 72%*

Class XII (CBSE) — 2021, 72% Class X (CBSE) — 2019, 60%

CERTIFICATIONS

- Google AI Essentials Specialization
- Google Python & Machine Learning
- Introduction to Programming Using Python
- Deloitte Australia — Technology Job Simulation
- Certificate of Participation — Dentsu AIgnite Assessment Round (Performance Marketing Track)
- Building Database Agent (Course Completion)

AWARDS & ACHIEVEMENTS

- 3rd Place — Inter-College Hackathon (2022): Developed a Bank Management System in C as part of a team.
- Currently developing an AI-powered calling agent for MIT ADT University under faculty guidance, automating student communication for admissions, courses, and campus services.
- All-rounder recognition in BCA.

ADDITIONAL INFORMATION

Languages: English, Hindi Willing to Relocate: Yes Availability: Immediate Joiner